SatsPath v2 Server Migration & Identifier Portability

Overview

A domain owner must be able to move from one SatsPath server to another while keeping `user@domain`, owner keys, histories, and verifier trust intact. Endpoint ownership and identity ownership are strictly decoupled.

Identity Layers

SatsPath separates five distinct identity layers:

Layer	Controls	Survives migration?
Namespace authority key	Domain policy, descriptor signing	Yes (held by domain owner)
Log identity (`log_id`)	Append-only history continuity	Yes (preferred: same log continues)
Log operator key	Checkpoint signing	Rotated via `OperatorKeyRotation`
Endpoint / TLS identity	Network reachability	Changed (new server URL)
Individual owner keys	User identity events	Always preserved

Migration Cases

1. Planned Move (Old Server Cooperating)

The old server exports a signed `MigrationStatement` binding its latest checkpoint to the new endpoint. The new server imports the full event stream, verifies it, and begins serving.

2. Old Server Offline / Censoring

If the domain owner holds the namespace authority key, they can issue a new `NamespaceDescriptor` pointing to the new endpoint. Clients with pinned checkpoints verify continuation via consistency proofs from the new server.

3. Operator Key Compromise

The namespace authority issues an `OperatorKeyRotation` and a new descriptor epoch. The old operator key is revoked.

4. Witness Set Change

A new descriptor epoch lists updated witness pubkeys and quorum requirements.

5. Replica Promotion

A healthy replica is promoted by updating DNS and issuing a migration statement.

6. Hosted User Moving to New Domain

A provider-owned domain cannot be unilaterally redirected. Instead, a signed redirect links the old `alice@provider.example` to a new sovereign identifier `alice@sovereign.example`. Clients display this as a **new identifier**, not a transparent migration.

Export / Import Format

A portable `MigrationExport` contains: - Signed `NamespaceDescriptor` (current epoch) - Full signed event stream (`Vec<NameEvent>`) - All checkpoints with operator signatures - Consistency proofs bridging checkpoints - Current-state map reconstruction data - Witness cosignature receipts - Operator key transition history

Exports contain **no identity private keys** unless a separate local client backup action handles them explicitly.

Security Properties

- A malicious old or new server cannot fabricate a migration.
- DNS rollback and endpoint rollback are detected and rejected.
- Clients with an old valid descriptor receive a precise `migration_required` or `policy_changed` result.
- The new server reconstructs and verifies the full state before advertising readiness.